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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Summer Examination - 2023</b><br><b>Course: S. Y. B. Tech.      Branch : Mechanical Engineering      Semester : IV</b><br><b>Subject Code &amp; Name: BTMPE405C Fluid Machinery</b><br><b>Max Marks: 60      Date:      Duration: 3 Hr.</b>                                                                   |                                                                                                                                                                                                                                                                                                   |                               |           |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly. |                                                                                                                                                                                                                                                                                                   |                               |           |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                   | (Level/CO)                    | Marks     |
| <b>Q.1</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                            |                               | <b>12</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>Obtain an expression for force exerted and work done by the jet of water on the curved plate when the plate is moving in the direction of the jet.</b>                                                                                                                                         | <b>Understanding/<br/>CO1</b> | <b>6</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>Water is flowing through a pipe at the end of which a nozzle is fitted. The diameter of the nozzle is 100 mm and the head of water at the center nozzle is 100 m. Find the force exerted by the jet of water on the fixed vertical plate. The coefficient of velocity is given as 0.95.</b>    | <b>Applying/CO1</b>           | <b>6</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>A jet of water of diameter 75 mm moving with a velocity of 25 m/s strikes a fixed plate in such a way that the angle between the jet and plate is 60°. Find the force exerted by the jet on the plate<br/>i) In the direction normal to the plate and<br/>ii) In the direction of the jet.</b> | <b>Applying/CO2</b>           | <b>6</b>  |
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| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                            |                               | <b>12</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>With a neat sketch explain construction and working of Francis turbine.</b>                                                                                                                                                                                                                    | <b>Understanding/<br/>CO2</b> | <b>6</b>  |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>Define the following terms<br/>i) Jet Ratio<br/>ii) Speed Ratio<br/>iii) Flow Ratio</b>                                                                                                                                                                                                        | <b>Understanding/<br/>CO3</b> | <b>6</b>  |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>Two jets strike the buckets of Pelton wheel, which is having shaft power as 15450 kW. The diameter of each jet is given as 200 mm. If the net head on the turbine is 400 m, find overall efficiency of the turbine. Take <math>C_v = 1.0</math>.</b>                                           | <b>Applying/CO4</b>           | <b>6</b>  |
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| <b>Q. 3</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |                               | <b>12</b> |
| <b>A)</b>   | <b>Which factor affects during the selection of a Hydraulic turbine?</b>                                                                                                                                                                                                                                                                                                                                                                                         | <b>Understanding/<br/>CO5</b> | <b>6</b>  |
| <b>B)</b>   | <b>What are the characteristics curves of a turbine? Explain main characteristics curve for reaction turbine.</b>                                                                                                                                                                                                                                                                                                                                                | <b>Understanding/<br/>CO3</b> | <b>6</b>  |
| <b>C)</b>   | <b>Enlist the names and explain the different safety devices in governing of turbines.</b>                                                                                                                                                                                                                                                                                                                                                                       | <b>Understanding/<br/>CO4</b> | <b>6</b>  |
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| <b>Q.4</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |                               | <b>12</b> |
| <b>A)</b>   | <b>What is priming? Why is it necessary?</b>                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Understanding/<br/>CO3</b> | <b>6</b>  |
| <b>B)</b>   | <b>What are the effects of cavitation? Give the necessary precaution against cavitation?</b>                                                                                                                                                                                                                                                                                                                                                                     | <b>Understanding/<br/>CO4</b> | <b>6</b>  |
| <b>C)</b>   | <b>The outer diameter of an impeller of a centrifugal pump is 400 mm and outlet width 50 mm. The pump is running at 800 r.p.m and is working against a total head of 15 m. The vanes angle at outlet is <math>40^\circ</math> and manometric efficiency is 75%. Determine;<br/>i) Velocity of flow at outlet,<br/>ii) Velocity of water leaving the vane, and<br/>iii) Angle made by the absolute velocity at outlet with the direction of motion at outlet.</b> | <b>Applying/CO6</b>           | <b>6</b>  |
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| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |                               | <b>12</b> |
| <b>A)</b>   | <b>Explain construction and working of vane type pump and give its advantages and disadvantages.</b>                                                                                                                                                                                                                                                                                                                                                             | <b>Understanding/<br/>CO6</b> | <b>6</b>  |
| <b>B)</b>   | <b>What are the steps to installation of pumps?</b>                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Understanding/<br/>CO7</b> | <b>6</b>  |
| <b>C)</b>   | <b>Classify special purpose pumps and write its applications.</b>                                                                                                                                                                                                                                                                                                                                                                                                | <b>Understanding/<br/>CO6</b> | <b>6</b>  |
|             | <b>*** End ***</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               |                               |           |

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| <b>Q.4</b>  | <b>Solve Any Two of the following.</b> <i>(This is just a sample instruction)</i> |            |          |
| <b>A)</b>   | Explain different types of casing used in centrifugal pumps.                      | <b>CO5</b> | <b>6</b> |
| <b>B)</b>   | Derive the expression for minimum speed for starting of Centrifugal Pump          | <b>CO6</b> | <b>6</b> |
| <b>C)</b>   | Explain necessity of priming and discuss methods of priming.                      | <b>CO5</b> | <b>6</b> |
|             |                                                                                   |            |          |
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b> <i>(This is just a sample instruction)</i> |            |          |
| <b>A)</b>   | With neat sketch explain working of Gear Pump and Lobe Pump.                      | <b>CO7</b> | <b>6</b> |
| <b>B)</b>   | Discuss the steps for installation of Efficient Pumps.                            | <b>CO7</b> | <b>6</b> |
| <b>C)</b>   | Explain the causes and remedies of failure of pumping systems.                    | <b>CO7</b> | <b>6</b> |
|             | <b>*** End ***</b>                                                                |            |          |

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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Regular &amp; Supplementary Summer 2024</b><br><b>Course: B. Tech.      Branch :Mechanical and Allied/ Productio / Automobile</b><br><b>Subject Code &amp; Name: BTMPE405C Fluid Machinery      Semester : IV</b><br><b>Max Marks: 60      Date: 24/06/2024      Duration: 3 Hrs.</b>                                                                                           |                                                                                                                                                                                                                                                                                                                                                                             |            |           |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly.<br>5. Illustrate your answers with sketches where ever necessary. |                                                                                                                                                                                                                                                                                                                                                                             |            |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                             | (Level/CO) | Marks     |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                      |            | <b>12</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            | Derive the expression for work done and efficiency for arrangement with water jet having velocity 'V' striking a flat plate moving with velocity 'u' moving in the direction of jet.                                                                                                                                                                                        | 2/CO1      | 6         |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            | A jet of water, diameter 70 mm strikes a curved vane at its centre with a velocity of 15m/sec, the curvd plate is moving with a velocity of 6m/sec in the direction of jet. If the blade is smooth and water is deflected by $165^\circ$ inside the blade, determine the following<br>(i) Normal force on blade<br>(ii) Efficiency of the system                            | 3/CO1      | 6         |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            | Derive the expression for Eulers equation of moment of momentum, state the assumptions made therein.                                                                                                                                                                                                                                                                        | 2/CO1      | 6         |
| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                      |            | <b>12</b> |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            | A Pelton wheel develops 10MW power while working under a head of 500m. The overall efficiency of turbine is 80%. If the jet ratio is limited to 12 and coefficient of velocity of nozzle is 0.98 and speed ratio is 0.47, determine the nozzle diameter, speed and the wheel diameter.                                                                                      | 3/CO2      | 6         |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            | Give reasons in brief (any three)<br>(i) Reaction turbine must run full of water.<br>(ii) Pelton turbines are called velocity turbines.<br>(iii) Kaplan turbines are best suited for part load operation.<br>(iv) Pelton turbines buckets are provided with double ellipsoids with a splitter edge<br>(v) Outlet triangle of a reaction turbine is a right angled triangle. | 3/CO2      | 6         |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            | Describe the Cavitation. How can it be avoided at the design stage and during operation? What are the likely places where it can occur and also list damages caused by Cavitation.                                                                                                                                                                                          | 2/CO2      | 6         |
| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                      |            | <b>12</b> |

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| A)          | What is governing and why it is necessary? Discuss with a schematic the governing of Pelton turbines.                                                                                                                                                                                                                                                                        | 2/CO3 | 6  |
| B)          | Describe and compare the main characteristics for Pelton and Francis turbines.                                                                                                                                                                                                                                                                                               | 2/CO3 | 6  |
| C)          | A turbine is to operate under a head of 25m at 200 rpm. The discharge is 9 m <sup>3</sup> /sec. If the turbine efficiency is 90%. If the head is reduced to 20m determine power generated, discharge, speed and specific speed. Also state the type of turbine.                                                                                                              | 2/CO3 | 6  |
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| Q.4         | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                                              |       | 12 |
| A)          | Explain as to how centrifugal pumps can be classified.                                                                                                                                                                                                                                                                                                                       | 2/CO4 | 6  |
| B)          | Describe various heads, losses and efficiencies of centrifugal pumps.                                                                                                                                                                                                                                                                                                        | 2/CO4 | 6  |
| C)          | Discuss about the following. (any three)<br>(i) Significance of duty point of a centrifugal pump<br>(ii) Centrifugal pumps arrangement in series and parallel<br>(iii) Suction lift of centrifugal pumps is limited to less than 25 feet<br>(iv) Priming and methods of priming<br>(v) Significance of Net positive suction head<br>(vi) Pumps are inefficient than Turbines | 2/CO4 | 6  |
|             |                                                                                                                                                                                                                                                                                                                                                                              |       |    |
| Q. 5        | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                                              |       | 12 |
| A)          | Describe the following special purpose pumps(any two)<br>(i) Pumps handling chemicals<br>(ii) Pumps used in nuclear power plant<br>(iii) Pumps to handle sewer slurries<br>(iv) Energy efficiency measures for pumps<br>(v) Submersible pumps                                                                                                                                | 3/CO5 | 6  |
| B)          | Illustrate the various centrifugal pump failures its causes and remedial action required to be taken by a maintenance engineer.                                                                                                                                                                                                                                              | 2/CO5 | 6  |
| C)          | Briefly illustrate the following positive displacement pumps (any two)<br>(i) Reciprocating pump<br>(ii) External Gear pump<br>(iii) Lobe pump<br>(iv) Sliding Vane pump                                                                                                                                                                                                     | 2/CO5 | 6  |
| *** End *** |                                                                                                                                                                                                                                                                                                                                                                              |       |    |